LEARNING TO IGNORE WHAT MATTERS: REPRESENTATION, PREDICTION, AND CONTROL IN JEPA WORLD MODELS

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ABSTRACT

A failed policy trained in a world model can reflect errors in the model, its task heads, or the controller-training procedure. We examine these alternatives in a jointly trained, squared-loss JEPA with a fixed 512-dimensional representation. In Craftax, one frozen model reaches 93.2% of 384 nearby goals with six open-loop actions and 96.1% with replanning, versus 35.7% for no-op; success remains 89.9% and 93.9% on displaced goals. A controlled moving-wall task then tests whether poor imagined control identifies a model failure. Across 45 matched PPO runs, replacing learned transitions, rewards, and terminations with exact simulator outcomes improves walls passed by 0.12–0.22 per episode across three model fits, but none of the 15 exact-environment policies exceeds 50% survival. Their mean survival remains below 1.5% for every model. The common controller is itself unsuccessful, so the earlier unmatched real-versus-imagined gap does not establish a task-solving deficit caused by learned transitions. A replicated nuisance–dynamics study supplies a separate representation-accessibility failure, while population constructions clarify the limits of non-collapse guarantees. Together, the measurements distinguish working local planning, unsuccessful policy training, and inaccessible state under a stated learning recipe.

1 INTRODUCTION

A world model can ignore texture while preserving a doorway's position, available opening, and response to action. Learning such abstractions without a downstream reward is a central attraction of joint-embedding predictive architectures (JEPAs): their targets are representations of future observations rather than future pixels (LeCun, 2022; Assran et al., 2023; Bardes et al., 2024). Recent systems demonstrate useful goal-conditioned planning from both pretrained and jointly learned representations (Assran et al., 2025; Sobal et al., 2025; Maes et al., 2026). The difficult question is what a particular success or failure establishes about the learned model.

We study this question through interventions on a jointly trained squared-loss JEPA. The model has a fixed 512-dimensional latent and a Gaussian marginal regularizer. The environments progress from Craftax to a moving-wall task and then a single cube with controlled nuisance and dynamics. Each step removes complexity to make a narrower question testable: can the model guide local actions, can its training environment support a competent controller, and under which conditions does its representation expose object position?

Craftax supplies a positive result with useful controls. On 384 fresh episodes, nearby position planning succeeds in 93.2% of open-loop trials and 96.1% with replanning. No-op succeeds in 35.7%, because some targets share the starting position; on the 247 displaced targets, planned success remains 89.9% and 93.9%. Goal generation and execution use independent randomness. These measurements establish local action-selection capability for the existing checkpoint.

The gate gives a different result. Earlier policies trained in imagination performed worse than policies trained on real transitions, and decoded predictions lost a visible wall. Those observations suggested a model limitation, but also changed the task heads and PPO protocol. We therefore train 45 policies under one matched protocol, crossing three frozen models and five policy seeds with learned transitions, exact transitions with fitted heads, and exact transitions with true signals. Even

the fully exact training environment yields mean survival below 1.5% for every model and no policy above 50%. Exact transitions and signals improve partial progress by 0.12–0.22 walls per episode, but the controller remains unsuccessful. This failure to qualify the common controller materially changes the interpretation of the original imagination result.

A separate replicated cube study exposes a representation-accessibility boundary. The same grid recipe supports subpixel position read-out when either noise is persistent or dynamics are deterministic, while per-frame noise with wind produces large error. Its training loss remains high, so the empirical failure is distinct from the known low-loss predictable-nuisance construction (Sobal et al., 2022). We retain that construction and an exact-Gaussian small-scale counterexample to state the objective's missing guarantees, rather than use them as explanations of the fitted networks.

The contribution is the measured scope of these comparisons: a positive local-planning capability, a matched intervention that prevents a stronger model-failure attribution, and a controlled nuisance–dynamics interaction in accessible state. Shared protocols, all final outcomes, and model versus policy replication make the evidence inspectable. The conclusions concern this learning recipe and its tested interfaces; closely related predictive-learning methods are discussed in Section 7.

2 Model and study design

Joint representation and prediction. An encoder maps an image to $z_t = f_\theta(x_t) \in \mathbb{R}^D$. An action-conditioned predictor uses H latent frames and their actions to predict the next latent:

$$\mathcal{L}(\theta) = \frac{1}{D}\widehat{\mathbb{E}}_t \left\| g_\theta(z_{t-H+1:t}, a_{t-H+1:t}) - z_{t+1} \right\|^2 + \lambda \,\mathrm{SIGReg}(\{z_t\}). \tag{1}$$

Gradients reach both the predictions and their targets; there is no stop-gradient or teacher. SIGReg compares the characteristic functions of random latent projections with those of a standard Gaussian (Balestriero & LeCun, 2025). It constrains the marginal distribution of features, not which physical variables they represent. The precise empirical and population expressions appear in Appendix B.

The main grid model uses a $4 \times 4 \times 32$ latent ($D = 512$), a tile-transformer encoder, and a factored spatial–temporal predictor with eight-frame history. Its representation training uses neither pixels as prediction targets nor rewards. Each model is fitted on $500{,}000$ transitions for approximately ten minutes on an A100; post-hoc decoders, heads, and policies are fitted separately. The Craftax and gate checkpoints use the same grid shape and training family but different data. Appendix D gives the complete settings and measured times.

Three tests. A frozen decoder tests whether information is *recoverable from an encoding* under additional supervision. Applying that same decoder to predicted latents tests whether it remains accessible in *model-generated futures*; because such latents can leave the encoder's distribution, a bad rendering alone does not prove that the information is absent. Finally, a controller evaluated in the real environment tests the *complete learned pipeline*, including its observation interface, reward and termination estimates, optimizer, and training distribution. These tests become more informative when their shared components and changed components are explicit.

3 Craftax: local planning works beyond staying still

The same question has a positive boundary in Craftax-Classic (Matthews et al., 2024): a compact JEPA can guide some real actions reliably. We evaluate the existing 512-dimensional checkpoint on 384 fresh starts across three evaluation seeds. After seven warm-up actions, six random actions generate a nearby target image. Goal generation and execution use independent simulator randomness. The planner uses terminal latent distance, without a decoder, reward head, privileged position input, or access to the generating actions. Success requires the target's player position after six real actions and survival throughout execution.

One-shot open-loop planning reaches $358/384$ targets (93.2%), and receding-horizon planning reaches $369/384$ (96.1%), against 16.9% for random actions and 35.7% for no-op (Figure 1). Because 137 targets share the starting position, displaced goals are reported separately: success remains 89.9% open loop and 93.9% with replanning. The goals are local, with displacements of zero

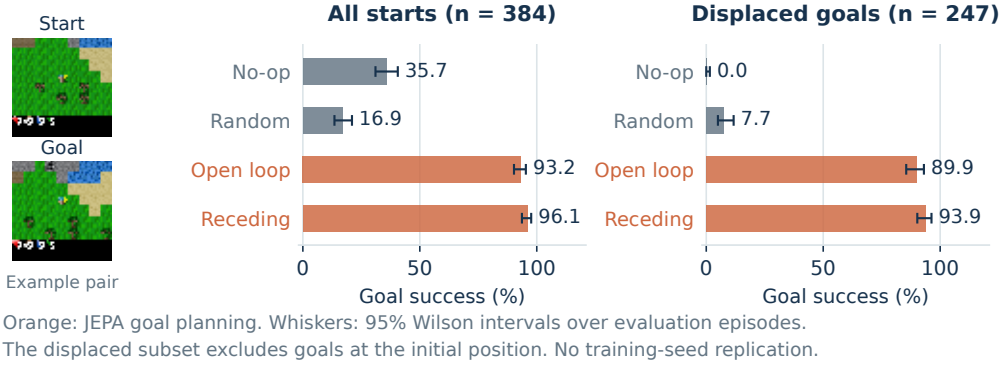


Figure 1: **Local Craftax planning under fresh execution noise.** One existing model is tested on 384 fresh episodes; 247 have a target different from the starting position. Whiskers are Wilson 95% intervals over evaluation episodes, conditional on this checkpoint. Receding-horizon planning uses more search compute than one-shot planning. The genuine start/goal pair is the first stored example with displacement at least two cells, selected without inspecting model outcomes.

to three grid cells. This establishes useful six-step position planning for one fitted model; evaluation seeds quantify fresh episodes, not independent model fits.

Replanning solves horizons $6, 5, 4, 3, 2, 1$, requiring 3.5 times the latent transition evaluations of the open-loop search. Its improvement therefore combines feedback and additional computation. These results concern local goal matching, not Craftax achievements or reward optimization. The earlier short reward-control experiment, including its weakly generalizing fitted heads, is retained separately in Appendix I.

4 The Gate: Separate Read-out Transfer from Control

A task with a delayed observation. A vertical wall approaches a controlled cube in a 63×63 image. Its gap is sampled at spawn, at the top or bottom with equal probability, and becomes visible at a fixed reveal point. From the centre, either gap needs five vertical moves but is revealed only two steps before contact; retreating creates time to respond. Wall or border contact terminates with reward -1, all other rewards are zero, and episodes last at most 128 steps. Three grid JEPAs are independently fitted to 500,000 uniform-random transitions, with a separate 25,000-transition evaluation archive.

The gate supports two complementary interventions on the same frozen models. First, we refit a read-out on forecast latents to test transfer from encoded observations. Second, we replace learned training transitions and task signals with exact simulator outcomes under a common PPO protocol. These interventions respectively test access to forecast state and the controller-training baseline.

4.1 Read-out Adaptation Lowers MAE without Reliable Position Tracking

The original decoder omits the wall from predicted latents within two steps for all three models, although it renders the wall from real encodings. To examine this transfer, we fit identical affine ridge read-outs on real future encodings, forecasts, or copied current encodings and evaluate their full cross-domain transfer. Training and evaluation use separate archives. Both predeclared train-only scaling schemes, four horizons, all models, and copied-current and constant baselines are retained in Appendix M.

Refitting reduces model 2's eight-step wall-position MAE from 16.47 to 9.11 pixels on the wall-present cohort. However, the forecast-fitted probe has RMSE 18.65 pixels, worse than 13.49 for the constant training-mean baseline. Its correlation with true position is only 0.246. In the prespecified

current-pre-reveal continuing slice ($n = 716$), refitting reduces MAE from 10.02 to 6.16 pixels, a paired gain of 3.85 $[3.48, 4.22]$. Yet its within-slice correlation is negative (-0.175). A post-hoc check using only the matching training cohort gives constant mean and median baselines of 4.77 and 4.64 pixels, both better than the adapted probe. The predeclared global-mean baseline gives 14.24 pixels on this slice, while the matching-cohort constants outperform the adapted probe. The adapted probe's advantage over a copied-current probe also has an interval containing zero. These measurements show changed read-out error, with a coarse association in one model; they do not demonstrate reliable position recovery. Models 0 and 1 have weaker real-latent controls and forecast-fitted errors near the constant baseline.

The forecast test therefore leaves the representation question unresolved at the precision needed for control. A missing decoded wall establishes failure of that decoder on forecasts. Refitting a state probe, by itself, does not turn a favorable conditional MAE into evidence of accurate dynamics.

4.2 The exact-environment baseline fails to solve the gate

None of the 15 policies trained with exact transitions and true task signals reaches 50% survival. Their mean survival is 0.47%, 0.39%, and 1.41% for model seeds 0, 1, and 2 (Figure 2; Table 13 in the appendix). The same controller is therefore unsuccessful even when transition, reward, and termination prediction errors are removed from its training environment. The poor policy performance in imagination therefore cannot be attributed to the learned transition model alone.

A fixed training protocol. For every frozen model, we train five recurrent MLP-PPO policies (Schulman et al., 2017) in each of three conditions: learned transitions with fitted reward/termination heads; exact simulator transitions re-encoded by the same encoder with the same fitted heads; and exact re-encoded transitions with true rewards and terminations. Actor inputs contain no simulator state. The 45 runs share their start-state mixture, context burn-in, rollout length, optimizer, updates, and evaluation keys within model/policy-seed pairs. Each completes 150 updates of 512×16 transitions and is evaluated at its final checkpoint on 256 real episodes. Walls passed counts exits before or on the terminal transition, including an exit on the same step as death. Survival requires remaining alive through step 128; these are the two primary outcomes; survival above 50% is a secondary operational threshold.

Exact-simulator trajectories become absorbing after true termination. With fitted heads, learned continuation may still weight those terminal states. We report this support change alongside the transition intervention. Policy-seed intervals are conditional on a fixed model and its evaluation set; the three model effects remain separate. Appendix K records the pairing checks, uncertainty estimands, and all contrasts.

Exact transitions and true signals nevertheless improve walls passed from 0.285 to 0.452, 0.217 to 0.341, and 0.320 to 0.543 in the three models. The paired gains are 0.166, 0.123, and 0.223 walls per episode. Survival changes are -0.47, -0.39, and $+1.02$ percentage points. Thus the substitution changes partial progress, while leaving task-level control poor in every fitted model. All 45 policies complete the prescribed budget; no intermediate policy or favorable model seed is selected.

4.3 Transition and task-signal substitutions have different effects

Keeping the fitted heads while substituting exact transitions lowers mean walls passed to 0.230, 0.068, and 0.096. Replacing those heads with true signals then raises the means to 0.452, 0.341, and 0.543. These contrasts expose sensitivity to how transitions and task signals are combined. They do not establish that less accurate dynamics are intrinsically better: the fitted-head condition applies its learned continuation after the simulator has entered an absorbing terminal state. Its logged post-terminal weight fraction averages 12.3%, 11.2%, and 9.4% across updates and policy seeds in the three models.

Real-transition head diagnostics already show a weak wall-collision signal. For model 0, fitted reward is about -0.05 on wall deaths and -0.89 on border deaths, against the true -1; termination AUC is 0.65 versus 0.96. Yet correcting both dynamics and signals still leaves the common MLP-PPO recipe unsuccessful. The exact condition leaves the actor's access to encoded state and its training procedure as unresolved limitations.

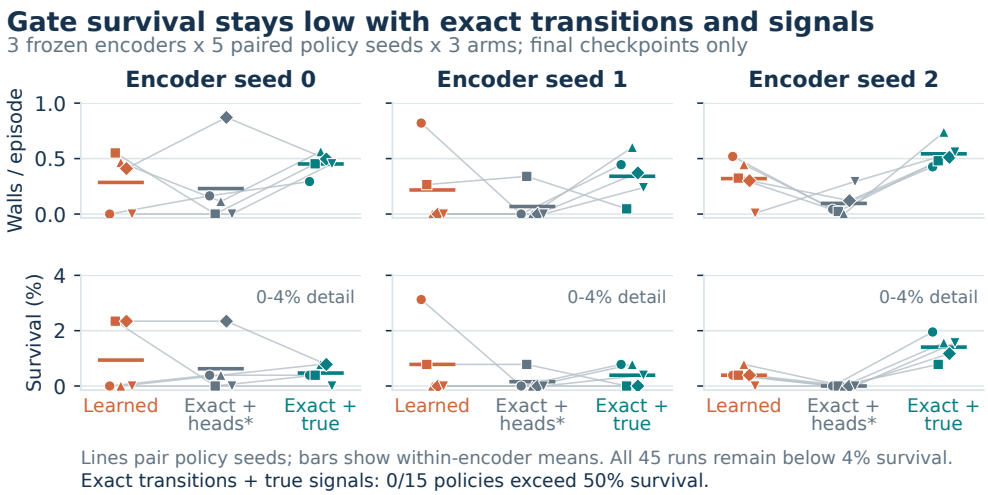


Figure 2: **The exact-environment control is itself unsuccessful.** All 45 final policies, with five paired policy seeds per model and training condition. Lines connect matched policy seeds; model fits remain separate. The survival panels use a 0–4% detail scale, far below the 50% operational threshold. Exact+heads* uses absorbing true terminal states while retaining fitted reward and continuation, so this transition substitution also changes terminal support.

4.4 A NATIVE SPATIAL POLICY ALSO LEAVES THE TASK UNSOLVED

A separate sensitivity replaces the MLP with a CNN on the native $4 \times 4 \times 32$ latent grid, without a decoder, extra latent coordinates, or privileged inputs. It repeats all three training conditions for three policy seeds per model, adding 27 complete runs. None exceeds 50% survival. With exact transitions and true signals, mean survival is 1.17%, 0.52%, and 0.65%; wall counts are 0.616, 0.367, and 0.452 (Appendix L). Thus changing this policy inductive bias does not qualify the short-rollout recipe as a successful control test.

The transition substitution has a different sign under this torso: exact transitions with fitted heads increase mean wall counts in all three models, whereas they decrease them in the primary MLP study. We retain the two campaigns separately. Across both, none of 72 policies crosses the threshold, including all 24 trained with exact transitions and true signals. Earlier real-environment CNNs on supervised decoded encodings do cross it in two of nine runs, using a different training protocol; those results establish a separate useful read-out, not a successful common baseline (Appendix J).

4.5 CORRECT TERMINATION REMOVES THE EXTENSION BUT DOES NOT RESOLVE CONTROL

The observed post-terminal weights trigger a separate sensitivity with exact transitions, true termination, and the original learned reward head. All 24 additional policies complete the fixed budget and have exactly zero continuation-weighted post-terminal mass. None exceeds 50% survival. Changing termination alone has mixed effects on walls passed and lowers the mean in all three native-grid cells (Appendix N).

With exact transitions and true termination held fixed, replacing the fitted reward with the true reward increases mean wall counts in all six model/torso cells: 0.287, 0.124, and 0.465 walls for the MLP, and 0.514, 0.367, and 0.271 for the native-grid CNN. Survival changes remain small and mixed. This identifies a contribution from the fitted reward interface within these training conditions, while the unsuccessful exact-condition controls leave encoder accessibility, policy interface, and training procedure unresolved. The separate horizon-16 campaigns total 96 policies with no threshold crossings, all conditioned on only three fitted world models.

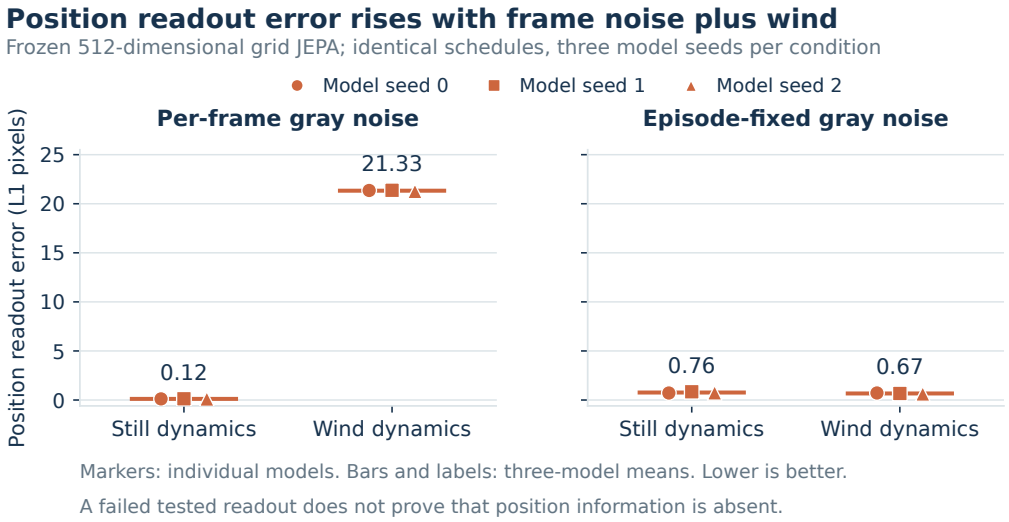


Figure 3: **The matched gray-noise cube comparison.** Position read-out from real encodings for three independently initialized grid JEPAs in each condition. Architecture, latent dimension, update count, and training dataset size are fixed. A high probe error is a failure of the tested read-out, not proof of absent information.

Table 1: **A matched nuisance–dynamics interaction.** Grid-JEPA position L1 error in pixels, mean $\pm$ sample standard deviation across three model fits. The same supervised read-out is applied to real encodings and eight-step predictions; the latter can also reflect read-out extrapolation. Predicted-window evaluation excludes resets and uses overlapping windows.

Gray background / dynamics	Real encoding	Eight-step prediction
Per-frame / deterministic	0.115 ± 0.012	4.84 ± 1.13
Per-frame / wind	21.330 ± 0.057	20.39 ± 0.92
Per-episode / deterministic	0.764 ± 0.068	1.83 ± 0.22
Per-episode / wind	0.668 ± 0.049	10.95 ± 0.53

5 Remove the wall: nuisance and dynamics interact

Cube worlds reduce the scene to one moving object. Random backgrounds either change independently each frame or remain fixed for an episode; dynamics are deterministic or include hidden wind. A fixed background is predictable but irrelevant to position. Per-frame noise is unpredictable and also irrelevant. These cases distinguish visual complexity from predictive usefulness.

The strongest replicated comparison holds the 512-dimensional grid architecture, dataset size (500,000 transitions), and training length (8,500 updates) fixed across four gray-noise conditions. A supervised nonlinear position read-out is fitted on training encodings and evaluated on independent evaluation frames. It achieves subpixel error whenever either the background is persistent or dynamics are deterministic. Combining per-frame noise with wind raises error to 21.33 ± 0.06 pixels across three model initializations (Table 1). This is a failure to expose position through the tested read-out. Its prediction loss remains high, about 0.73 per coordinate, so it is not evidence of a successfully optimized low-loss nuisance solution.

Other fitted models fail at different interfaces. On a black background without wind, the class-token JEPA permits 0.087-pixel position read-out from real encodings, but that read-out has 45.27-pixel error on eight-step predictions. A flow model also fails on persistent color texture, with 18.56-pixel real-latent error. These are single-fit observations with different dataset sizes, reported separately in Appendix C; they preclude a uniform ranking of model families. Together, the cube results motivate

an objective-level question rather than a universal empirical mechanism: what does low prediction loss plus non-collapse actually guarantee?

6 NON-COLLAPSE LEAVES FEATURE CHOICE OPEN

A Gaussian marginal can describe either useful state or an episode-specific nuisance. The known static-nuisance construction of Sobal et al. (2022) makes this explicit: an encoder can Gaussianize an irrelevant constant texture and a predictor can copy it, obtaining zero population prediction loss and zero Gaussian discrepancy. We state its control consequence and assumptions in Appendix A. This is a boundary of the objective's guarantees, not an identified explanation of the high-loss gray-frame/wind fits.

A second construction separates information from geometric scale. An exactly Gaussian latent can encode a control-relevant bit invertibly while placing its two values arbitrarily close together. For an L-Lipschitz downstream score, latent root-mean-squared prediction error bounds score error only after multiplication by L; exact decodability supplies no bound on that sensitivity. The construction uses unrestricted measurable coding and a fresh unpredictable bit. Its full statement and proof clarify this narrow scope, without turning a failed neural read-out into a proof that information has been deleted.

7 RELATED WORK

Which JEPA setting? Image and video JEPAs learn by predicting representations (Assran et al., 2023; Bardes et al., 2024). DINO-WM fits dynamics on frozen visual features, and V-JEPA 2 freezes its encoder for action-conditioned prediction with an L1 objective (Zhou et al., 2024; Assran et al., 2025). PLDM instead jointly learns its encoder and dynamics with multiple regularizers (Sobal et al., 2025). Our direct setting is LeWorldModel's jointly trained squared-loss predictor with SIGReg (Maes et al., 2026; Balestriero & LeCun, 2025); the population statements do not automatically transfer to these other objectives.

Predictability, geometry, and planning. Predictable nuisance is an established limitation of joint embeddings (Sobal et al., 2022). Recent work studies how contrastive sampling changes slow-noise bias (Gulati & Nemenman, 2026), and how temporal-distance objectives improve planning-relevant geometry (Bai & Xiong, 2026). LeWorldModel already reports accurate probes alongside weaker planning, while Terver et al. (2026) examine architecture and planner choices and discuss multi-modal conditional means. Self-predictive abstraction analyses similarly distinguish predictive structure from task sufficiency (Tang et al., 2023; Ni et al., 2024). Physical-state grounding also directly tests recoverability and goal-conditioned control, using privileged state or proprioceptive supervision beyond a prediction-only objective (Yan et al., 2026; Liu et al., 2026). These precedents motivate a controlled intervention in a specified learning recipe, rather than a first claim that prediction and control can disagree.

Models for decisions. Prediction–control mismatch predates JEPAs (Lambert et al., 2020). Value-aware model learning, value equivalence, and proper value equivalence ask which errors affect downstream Bellman computations and policy values (Farahmand et al., 2017; Grimm et al., 2020; 2021). Expectation models can suffice for suitable value-function classes (Wan et al., 2019); conditional-mean prediction is therefore not inherently incompatible with control. Stochastic imagination models provide another route (Hafner et al., 2020; Alonso et al., 2024; Hafner et al., 2025), but they too can fail at encoding, action conditioning, or rollout stages (Hansen & Wang, 2026). Temporal-difference JEPA instead predicts policy-conditioned successor features with detached TD targets and EMA target networks, allowing value-based control in stochastic MDPs (Bagatella et al., 2026). RC-aux retains target gradients and adds multi-horizon prediction and reachability supervision in a deterministic-dynamics analysis (Li et al., 2026). These objectives provide stronger or different constraints than one-step squared prediction with a marginal prior.

8 DISCUSSION

The most consequential gate result is the unsuccessful exact-environment control. Removing learned transition and task-signal errors improves wall counts, yet every fitted model remains below 1.5% mean survival with the common MLP-PPO recipe. An explanation based only on poor world-model dynamics is therefore incomplete. The earlier decoder/CNN result establishes that at least one frozen encoder contains useful task information under a different read-out and training procedure; it does not supply the missing successful baseline for this matched controller.

Craftax and the nuisance cubes define complementary boundaries. The existing JEPA guides local position-reaching actions beyond a no-op baseline under independent execution noise. The same model family has a reproducible position-readout failure in the gray-frame/wind cube condition. Together, these observations support evaluating a proposed model improvement at the interface it actually changes: representation accessibility, forecast read-out, fitted task signals, or policy training.

The main limitations are model replication and identification. Craftax uses one checkpoint; the gate and replicated cube comparison use three. The exact-transition/fitted-head intervention changes terminal support, and none of the primary gate conditions learns reliable survival. The population theory addresses non-collapse and geometric scale under explicit measurable-function assumptions, with no finite-network or optimization guarantee. It does not identify the mechanism of the empirical cube failure. These boundaries determine which conclusions can be drawn from the completed results.

REPRODUCIBILITY STATEMENT

Appendices D–I specify model fitting, post-hoc supervision, evaluation protocols, and replication levels. The tables retain final-checkpoint outcomes rather than selecting intermediate peaks. The accompanying evidence manifest identifies the source artifacts used for each result; independent policy seeds are distinguished from independent model fits. Claims based on a single checkpoint are labeled throughout.

STATEMENT ON THE USE OF AI

Language models assisted with prose, code, plotting, proof checking, literature review, and adversarial manuscript review. The authors are responsible for all claims, experiments, and verification of the resulting text and artifacts.

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A WHAT THE OBJECTIVE DOES NOT GUARANTEE

The experiments concern fitted finite networks. The following population statements ask a narrower question: which guarantees follow from prediction error and a Gaussian latent marginal alone? They do not characterize optimization or identify the cause of the empirical failures.

A.1 NON-COLLAPSE DOES NOT SELECT THE CONTROLLED STATE

Write observations as $x_t = R(p_t, b)$, where p_t is controlled state and b is an episode-constant nuisance recoverable from the image. Assume that b is independent of states and actions in the training data. At deployment, assume it is independent of the initial state and transition randomness, and does not directly affect dynamics or reward. Let SIGReg_∞ be the nonnegative population characteristic-function discrepancy, which vanishes for $\mathcal{N}(0, I_D)$ (Appendix B).

Proposition 1 (Gaussian nuisance optimum; extending Sobal et al. (2022))**.** *If the encoder class contains a map $f(R(p, b)) = \varphi(b)$ with $\varphi(b) \sim \mathcal{N}(0, I_D)$ and the predictor class contains the last-latent copy map, their population prediction loss and SIGReg_∞ are both zero. This is a global optimum with covariance I_D and $I(z_t; p_{1:T}) = 0$ under the training distribution. A deployed controller observing only these latents, time, and its own actions has the value of a randomized state-blind policy.*

The argument is simple: nuisance changes across episodes, satisfying the marginal constraint, but is constant within each episode, making prediction exact. Its deployment value follows by giving a state-blind policy the same private nuisance draw and controller randomness. The Gaussianization assumption requires a rich nuisance and a realizable measurable map; a finite texture does not literally instantiate it. The finite-batch regularizer also differs from the population functional. A complementary innovation bound in Appendix B shows that an encoder preserving all state distances at a fixed scale must incur prediction error for unpredictable state innovations.

A.2 DECODABILITY DOES NOT SET A USEFUL PREDICTION SCALE

Even preserving all state information leaves its geometry unconstrained. For any $\varepsilon > 0$, we can place the two values of a relevant bit less than 2ε apart while keeping the *entire* latent distribution exactly standard Gaussian.

Proposition 2 (Exact information at arbitrarily small scale)**.** *For every $\varepsilon > 0$ and $D \geq 1$, there is a measurable observation process with episode-constant nuisance b and a fresh independent fair state bit p_t, and an encoder f_ε, such that $z_t \sim \mathcal{N}(0, I_D)$ and both p_t and b are exactly recoverable almost surely. For fixed b, $0 < \|f_\varepsilon(R(1, b)) - f_\varepsilon(R(0, b))\| < 2\varepsilon$. The optimal next-latent predictor is independent of the next bit and has coordinate-mean squared error in $(0, \varepsilon^2/D]$. With reward $\mathbf{1}\{a_t = p_t\}$, latent feedback attains value 1 per step, while every state-blind policy attains $1/2$.*

The construction partitions one Gaussian coordinate into narrow intervals and uses the bit to select the lower or upper half of each interval's conditional probability mass. The nuisance fixes the interval and the within-half quantile. This preserves the Gaussian marginal and exact decoding while making the two states arbitrarily close. The full proof is in Appendix B. It uses infinite-precision measurable coding; the next bit is intrinsically unpredictable. It therefore separates information from scale, without proving avoidable prediction error or explaining deletion of the predictable gate wall.

Why scale matters for a downstream read-out. For any fixed L-Lipschitz downstream score q (for example, an action-value read-out) and any coupled real and predicted next latents,

$$\left|\mathbb{E}q(Z') - \mathbb{E}q(\widehat{Z}')\right| \leq L\sqrt{\mathbb{E}\|Z' - \widehat{Z}'\|^2}. \tag{2}$$

This standard bound follows from Lipschitz continuity and Cauchy–Schwarz. Prediction error controls score error only relative to the read-out's sensitivity. Exact decodability alone supplies no bound on that sensitivity. We do not estimate L or claim a control guarantee for the fitted networks; the inequality makes explicit the missing bridge between a geometric loss and a decision.

B PROOFS

B.1 EMPIRICAL OBJECTIVE AND POPULATION IDEALIZATION

For a training batch of B windows and H predicted time positions, the implemented loss is

$$\mathcal{L}_{\text{pred}} = \frac{1}{BHD} \sum_{b,h} \|\widehat{z}_{b,h+1} - z_{b,h+1}\|^2. \tag{3}$$

The notation in Equation 1 averages over time and uses this coordinate-mean convention. One-step targets are teacher-forced at every temporal position, with a shorter history near the start of a window. For encoded windows of T frames, P random unit projection directions u_p, K knots $\tau_k \in [0, 3]$, and $w_k = e^{-\tau_k^2/2}$, the regularizer is

$$R = \frac{B}{TP} \sum_{t,p,k} q_k w_k \left[\left(\frac{1}{B} \sum_b \cos(\tau_k u_p^\top z_{b,t}) - w_k \right)^2 + \left(\frac{1}{B} \sum_b \sin(\tau_k u_p^\top z_{b,t}) \right)^2 \right], \tag{4}$$

where $\Delta = 3/(K-1)$, with $q_k = \Delta$ at the endpoints and 2Δ in the interior. The characteristic function is averaged across batch examples separately at each temporal position; the factor B is part of the implementation with $\lambda = 0.09$.

The population statements use the nonnegative discrepancy

$$\text{SIGReg}_\infty(\mathcal{L}(z)) = \int_{\mathbb{S}^{D-1}} \int_0^3 w(\tau) \left[\left(\mathbb{E}\cos(\tau\langle z, u\rangle) - w(\tau) \right)^2 + \left(\mathbb{E}\sin(\tau\langle z, u\rangle) \right)^2 \right] d\tau \, d\sigma(u), \tag{5}$$

where $w(\tau) = e^{-\tau^2/2}$ and σ is uniform on the sphere. Only the fact that this functional vanishes for a standard Gaussian is needed. It is not a theorem about exact minima of the finite-batch objective.

B.2 NUISANCE OPTIMUM AND THE COST OF INNOVATIONS

Proposition 1.

Proof. Let β recover the nuisance from the observation, set $f^\star = \varphi \circ \beta$, and let $g^\star$ copy the last latent. Then $z_t = \varphi(b) = z_{t+1}$, so every prediction residual vanishes. The Gaussian marginal gives covariance I_D and zero population regularizer. Both objective terms are nonnegative, so their sum is globally minimized at zero. Training-distribution independence gives $I(\varphi(b); p_{1:T}) = 0$.

For deployment, draw nuisance B and the controller's random seed U at reset, independently of the initial state and transition randomness. Run the original controller in the environment with nuisance B. Let a state-blind policy with private seed (B, U) feed constant latent history $\varphi(B)$ to a copy of that controller and reproduce its actions with U. Coupled to the same initial state and transition randomness, the policies take the same first action. If their actions agree through step t, their states, supplied latent histories, and seeds agree, so the next actions agree as well. Induction gives identical trajectories, returns, and terminations. This does not assume independence between B and states subsequently influenced by its induced actions. □

Corollary 1 (Innovation cost for uniformly state-resolving encoders)**.** *Let $p_{t+1} = p_t + \mu(a_t) + w_t$, $\mathcal{H}_t = \sigma(p_{\le t}, b, a_{\le t})$, and $\Sigma_t = \text{Cov}(w_t \mid \mathcal{H}_t)$. Assume finite second moments of states and encoded observations, $x_t = R(p_t, b)$, fixed nuisance b, and $\|f(R(p, b)) - f(R(p', b))\| \ge \kappa \|p - p'\|$ for all admissible p, p', b, with $\kappa > 0$. Every predictor of latent and action history satisfies*

$$\frac{1}{D}\mathbb{E}\|g - f(x_{t+1})\|^2 \ge \frac{\kappa^2}{D}\mathbb{E}\text{tr}\Sigma_t. \tag{6}$$

When $\mathbb{E}\text{tr}\Sigma_t > 0$, this exceeds the nuisance encoder's zero prediction loss under the assumptions of Proposition 1.

The assumption is stronger than injectivity or task sufficiency: every state distance must be preserved at a uniform scale. Innovation is measured after conditioning on state history and nuisance; persistent wind is not automatically fresh innovation.

Corollary 1.

Proof. Put $Z = f(x_{t+1})$ and $\mathcal{C}_t = \sigma(z_{\le t}, a_{\le t})$. Because observations and latents are measurable functions of state and nuisance, $\mathcal{C}_t \subseteq \mathcal{H}_t$. Any predictor using a shorter window is also $\mathcal{C}_t$-measurable. Conditional least squares and the tower property give

$$\begin{aligned}\mathbb{E}\|g - Z\|^2 &\ge \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &= \mathbb{E}\|Z - \mathbb{E}[Z \mid \mathcal{H}_t]\|^2 + \mathbb{E}\|\mathbb{E}[Z \mid \mathcal{H}_t] - \mathbb{E}[Z \mid \mathcal{C}_t]\|^2 \\ &\ge \mathbb{E}\mathrm{trCov}(Z \mid \mathcal{H}_t).\end{aligned}$$

The cross term vanishes because its first factor has conditional mean zero given $\mathcal{H}_t$. Take conditionally independent copies P, P' of p_{t+1} given $\mathcal{H}_t$. Such copies exist for Euclidean state spaces, using their regular conditional distributions. The nuisance b and current state p_t are fixed under this conditioning. The conditional independent-copy variance identity yields

$$\begin{aligned}\mathrm{trCov}(Z \mid \mathcal{H}_t) &= \tfrac{1}{2}\mathbb{E}[\|f(R(P,b)) - f(R(P',b))\|^2 \mid \mathcal{H}_t] \\ &\ge \tfrac{\kappa^2}{2}\mathbb{E}[\|P - P'\|^2 \mid \mathcal{H}_t] \\ &= \kappa^2 \mathrm{trCov}(p_{t+1} \mid \mathcal{H}_t) = \kappa^2 \mathrm{tr}\Sigma_t.\end{aligned}$$

The final equality follows because $p_t + \mu(a_t)$ is $\mathcal{H}_t$-measurable. Taking expectations and dividing by D proves Equation 6. Averaging these inequalities gives the corresponding bound for a multi-time prediction objective. Proposition 1 supplies the zero-loss comparator when its assumptions also hold. □

B.3 Conditional prediction and geometric scale

Lemma 1 (Conditional mean)**.** *For a fixed encoder and data law with $\mathbb{E}\|z'\|^2 < \infty$, the unique squared-error minimizer over measurable square-integrable predictors from history c and action a is $m(c,a) = \mathbb{E}[z' \mid c, a]$, up to null sets.*

Proof. Let $m = \mathbb{E}[z' \mid c, a]$. For every square-integrable measurable $g(c,a)$, expand $g - z' = (g - m) + (m - z')$. The cross term has expectation zero since $\mathbb{E}[m - z' \mid c, a] = 0$. Hence

$$\mathbb{E}\|g - z'\|^2 = \mathbb{E}\|g - m\|^2 + \mathbb{E}\mathrm{trCov}(z' \mid c, a).$$

The second term does not depend on g. The first is nonnegative and vanishes exactly when $g = m$ almost surely. □

The finite-sample regularizer. The finite-sample regularizer differs from Equation 5. For independent identically distributed samples with characteristic function ϕ and empirical estimate $\widehat{\phi}_n$,

$$\mathbb{E}|\widehat{\phi}_n(t) - w(t)|^2 = |\phi(t) - w(t)|^2 + \frac{1 - |\phi(t)|^2}{n}. \tag{7}$$

Indeed, $\mathbb{E}\widehat{\phi}_n = \phi$ and $\mathbb{E}|\widehat{\phi}_n - \phi|^2 = (1 - |\phi|^2)/n$, since each complex exponential has modulus one; expanding around ϕ gives the identity. At a Gaussian law, the residual equals $(1 - e^{-t^2})/n$. This is estimator bias, not an optimization floor. Changing the law changes both terms; dependent temporal samples introduce additional covariance terms. The implementation multiplies this statistic by batch size B, so its Gaussian finite-batch bias does not shrink as $1/B$ after that scaling. Finite projections and knots also do not characterize Gaussianity.

Proposition 2.

Proof. Partition $\mathbb{R}$ into $I_j = [2\varepsilon j, 2\varepsilon(j+1))$, $j \in \mathbb{Z}$. Let $q_j = \Pr(G \in I_j)$ for $G \sim \mathcal{N}(0,1)$. Every q_j is positive and their sum is one. Let F_j be the conditional Gaussian distribution function on I_j, with inverse quantile function Q_j on $(0,1)$. The conditional CDF is continuous and strictly increasing on the interval interior, so its inverse is measurable and takes values in that interior.

Draw J with probabilities q_j, $U \sim \mathrm{Uniform}(0,1)$, and $V \sim \mathcal{N}(0, I_{D-1})$, independently. Set $b = (J, U, V)$; for $D = 1$, omit V. Take $R(p, b) = (p, b)$ and define

$$f_\varepsilon(R(p,b)) = \big(Q_J((U+p)/2), V\big). \tag{8}$$

The mixture $(U + p_t)/2$ is uniform on $(0, 1)$, since p_t is an independent fair bit. Conditional on $J = j$, the first coordinate therefore has the Gaussian law restricted to I_j. Mixing with weights q_j recovers $\mathcal{N}(0, 1)$. Independence of V gives the joint law $\mathcal{N}(0, I_D)$ and zero population regularizer.

Given the first coordinate y, its interval determines J almost surely. Write $v = F_J(y)$. Then $p_t = \mathbf{1}\{v > 1/2\}$ and $U = 2v - p_t$ almost surely; the other coordinates recover V. Thus the latent recovers both state and nuisance. For fixed b, the two first coordinates are distinct and lie inside the same interval of width 2ε. All remaining coordinates agree, proving the distance bound.

Latent history reveals b, while the next bit remains independent and fair. The next latent consequently has two equiprobable values, say $z^{(0)}(b)$ and $z^{(1)}(b)$. Lemma 1 gives $m(b) = (z^{(0)}(b) + z^{(1)}(b))/2$. Its conditional squared error is $\|z^{(1)}(b) - z^{(0)}(b)\|^2/4$, strictly positive and bounded by ε^2. The midpoint is square-integrable by Jensen's inequality, since both state encodings have finite second moments. Taking expectations and dividing by D proves the coordinate-mean loss bound, including strict positivity since the conditional error is positive almost surely. The midpoint depends only on b, so it is independent of p_{t+1}. Finally, decoding the current bit permits $a_t = p_t$. A state-blind action is independent of the current fair bit, giving success probability $1/2$. □

C THE CUBE WORLDS

The position-readout study crosses black, per-frame random, and per-episode random backgrounds with deterministic or hidden-wind dynamics. Color-noise and black-background datasets contain 100,000 transitions; the gray-noise conditions contain 500,000. Only the four gray grid-JEPA conditions have three independent model initializations. Table 1 and Figure 3 report that replicated comparison. Tables 2 and 3 retain seed-0 outcomes for the broader conditions, using the same canonical benchmark files throughout. Differences across dataset sizes and model recipes are not isolated nuisance effects.

Environment. The scene contains a red 6×6 object in a 63×63 RGB image with a red boundary. Four cardinal actions move the object three pixels. Reset positions are uniformly sampled from integer coordinates in $[16, 42]^2$, and initial wind is zero. In wind conditions, each wind coordinate independently increments by -1, 0, or 1 with probabilities 0.1, 0.8, and 0.1, clipped to $[-2, 2]$; the updated wind is added to the action displacement. Background pixels are uniform integers from 0 to 255; gray noise shares the same sample across RGB channels. Contact with the boundary gives reward -1 and terminates; other rewards are zero, with a 128-step cap.

Read-out fitting. The nonlinear latent position probe is fitted on training data and evaluated on independent `eval.npz` frames. Its spatial softmax-and-offset output is produced by two hidden layers of width 128, trained for 10,000 updates with batch 2,048 and Adam at initial learning rate 10^{-3} with cosine decay. Grid inputs are standardized using training statistics. For class-token inputs, standardization and PCA whitening are selected using the final 10% of fitting rows, then the chosen variant is refitted on the full fitting split. Whitening floors singular values at 1% of the largest. Applying this read-out to predicted latents tests both prediction and read-out extrapolation. The pixel read-out used for decoded flow images is a separate supervised CNN trained with synthetic background replacement using ground-truth cube masks; it is a diagnostic, not a policy or a world-model training signal. Rollout windows are sampled with replacement and overlap, and exclude death or reset during the eight context and 16 forecast frames. The configured benchmark uses 2,048 windows for video metrics and the first 512 for position diagnostics; exact sampled indices were not retained in the archived result files. Surviving windows are a selected subset of wind trajectories, so these errors do not establish death prediction or closed-loop control.

Earlier controller experiments use the three black/color background conditions. Their final survival outcomes are retained below. The class-token JEPA's near-zero loss on per-episode texture and poor reward-driven control are consistent with the known predictable-nuisance mechanism, without exactly realizing Proposition 1. The replicated gray-frame/wind failure instead has high prediction

Table 2: Seed-0 cube position L1 error (pixels) for the grid and class-token JEPAs. "Real" evaluates held-out encoded frames; "Pred. 8" evaluates the same read-out on eight-step predicted latents. Gray conditions use more training transitions than black and color conditions.

Background	Wind	Grid JEPA		Class-token JEPA	
		Real	Pred. 8	Real	Pred. 8
Black	Off	0.097	3.933	0.087	45.268
	On	0.126	11.732	0.049	8.957
Color, frame	Off	0.172	5.570	24.133	19.428
	On	0.253	11.823	25.185	19.317
Color, episode	Off	3.273	6.594	20.804	21.567
	On	1.451	13.342	23.431	21.807
Gray, frame	Off	0.115	4.269	23.954	20.286
	On	21.354	19.696	25.288	19.567
Gray, episode	Off	0.715	1.779	21.316	21.284
	On	0.717	11.553	23.880	23.386

Table 3: Seed-0 flow cube position L1 error (pixels). Latent and pixel columns use different supervised read-outs. The persistent-color failure is visible before dynamics: real encoded position has large error. These are model-recipe references rather than objective-only comparisons.

Background	Wind	Real latent	Pred. 8 latent	Pred. 8 pixels
Black	Off	0.140	0.126	0.242
	On	0.153	11.632	11.622
Color, frame	Off	1.088	1.059	0.395
	On	1.344	12.311	11.985
Color, episode	Off	18.563	21.954	22.423
	On	7.010	18.960	16.937
Gray, frame	Off	3.114	2.667	0.382
	On	3.690	14.392	12.302
Gray, episode	Off	5.062	4.896	1.339
	On	3.066	15.150	13.583

loss and should not be interpreted as the same mechanism. Still-world survival can be achieved by state-blind alternating actions, so it does not certify state feedback.

Table 4: Survival (percent) in the supplementary cube worlds. "LeWM" denotes the class-token JEPA; "grid" denotes the grid JEPA; "goal" and "heads" use latent-goal and learned-return search, respectively. PPO columns use the final saved policies; "px" and "z" identify pixel and latent observations. Still-world survival does not certify state feedback.

Background	Black		Per-frame noise		Per-episode noise	
Dynamics	Still	Wind	Still	Wind	Still	Wind
random	20	0	20	0	20	0
centre	100	99	100	99	100	99
LeWM goal	97	64	5	0	80	0
LeWM heads	100	60	9	0	2	0
LeWM PPO	100	99	0	0	0	0
grid goal	88	1	92	0	100	77
grid heads	100	5	100	0	85	8
grid PPO	100	94	100	0	2	0
flow PPO px	100	94	98	97	98	95
flow PPO z	100	96	99	1	51	1
PPO real	100	98	100	96	100	78

D HYPERPARAMETERS AND COMPUTE

The gate and gray-noise cube conditions supply 500,000 uniform-random transitions each. The earlier black and color-noise cube datasets contain 100,000 transitions. Craftax uses its separate 500,000-transition archive. Model training uses bfloat16 compute, float32 parameters, and one A100. A 500-step calibration estimates the training length fitting 575 seconds including compilation; the grid gate model trains for 8,500 steps (17,000 and 34,000 in the earlier budget extensions of Appendix J), the class-token cube models for 23,750–24,000. Model seeds 0, 1, 2 of the grid JEPA differ only in the initialisation and data-order seed.

JEPA settings. Both JEPA architectures use batch 128 and AdamW with learning rate and weight decay 10^{-3}, 10% warm-up followed by cosine decay to zero, and gradient clipping at 1.0. SIGReg uses weight 0.09, 17 knots on $[0, 3]$, and 1024 random projections. Neither architecture uses a stop-gradient, teacher, decoder, or reward during world-model fitting. Linear–BatchNorm–GELU–linear projection heads flatten batch, time, and spatial tokens for normalization. Training-mode statistics therefore couple examples and temporal positions; at inference, running statistics define a fixed pointwise projection and the attention predictor is causal. The theoretical causal-map formulation is an inference/population idealization.

	Class-token JEPA	Grid JEPA
encoder	ViT, 7×7 patches	tile transformer + conv
width / depth / heads	192 / 6 / 3	128 / 6 / 8
encoder MLP	768	$4\times$ width
read-out	class token	4×4 grid, 32 channels
latent dimension	192	512
projector	BN projector, width 2048	same, hidden 512
predictor	AdaLN-zero AR	spatial + causal temporal
predictor depth / heads / dim	6 / 16 / 64	6 / 8 / 32
predictor MLP / width	2048 / latent	$2\times$ width / 256
history	3 frames	8 frames
dropout	0.1	0
parameters	15.3M	2.6M + 8.3M

Flow settings. The tokenizer uses 9,000 steps, batch 256, and learning rate 9×10^{-4}; dynamics 16,000 steps, batch 64, the same learning rate, width 256, six layers, eight spatial and temporal attention heads, a 32-frame context, independent per-frame noise levels, a shortcut loss of weight 0.25 every fourth step after the first quarter of training, history dropout 0.25, two history frames resampled every eight steps after half of training, and a temporal loss weight of 0.25. The symlog reward head has 255 bins over $[-20, 20]$; reward and termination losses have weight 0.1 and enter the trunk at full strength for half of training, then at factor 0.1. Inference uses four function evaluations per frame. We corrected terminal-transition masking in the recipe's reward loss (Appendix H).

JEPA read-outs. Gate post-hoc heads use the frozen encoder and the first 800 time rows (80,000 frames) of the same archive used for world-model training. The following 200 time rows are held out from head fitting only; the independent world-model evaluation archive is `eval.npz`. The pixel decoder uses 8,000 steps, batch 256, learning rate 10^{-3}, width 256, six residual blocks; it supports the figures and the decoded-latent control, in which its frozen output, clipped to $[0, 255]$, replaces the pixels of the real-gate PPO; no true pixel enters that policy after the encoder. Reward, termination, and behaviour-value heads are two-layer perceptrons of width 512 on the latent and the one-hot action, 6,000 steps, batch 1024, learning rate 10^{-3}, Monte Carlo value targets at discount 0.99, value loss weight 0.1. The residual predictor models $z_{t+1} - m$ with $m = g(z_{t-7:t}, a_{t-7:t})$ the original grid prediction, conditioned on m, z_t, and the action: three residual blocks of width 1024, 8,000 steps, and batch 2048. The final 20,000 valid windows of the same world-model training archive are excluded from residual fitting and used for its quality check. They are not independent world-model evaluation data. For encoder seed 0, coordinate-mean MSE is 3.07×10^{-4} for the original predictor, 2.05×10^{-4} for one residual sample, and 1.32×10^{-4} for the mean of eight. This

quality check uses eight function evaluations per sample; the reported PPO runs instead use four. Averaging eight learned samples is a random Monte Carlo average, not an exact conditional mean.

Policy optimisation. PPO uses discount 0.99, GAE parameter 0.9, clipping 0.2, value coefficient 0.5, entropy coefficient 0.01, AdamW at 2.5×10^{-4} with weight decay 0.01, four epochs per batch, and gradient clipping at 1.0. Pixels pass through a three-block Impala torso (Espeholt et al., 2018), latents through a two-layer perceptron; a GRU receives that embedding and the previous action; the critic uses 64 categorical bins with histogram-Gaussian targets (Farebrother et al., 2024). Real-environment PPO uses 128 environments and 64 steps per iteration with four minibatch groups, truncated backpropagation over 16 steps, and 1.5M environment steps. In-model PPO uses 150 updates of 512 trajectories of 16 steps (1.2M imagined steps) with minibatch size 128; half of the starts are resets, half dataset states with their real eight-frame context. All reported policy rows use final checkpoints.

Compute. On a shared A100, the seed-0 grid-gate JEPA completes 8,500 updates in 573 seconds and has a logged total of 586.5 seconds. Probe and head fitting takes approximately three minutes. The residual sampler's optimization stage takes 74.3 seconds; model loading, latent extraction, fitting, and quality evaluation together take 275.2 seconds before the final file write. Other logged stages are PPO inside the JEPA 2.0–4.3 min per run; PPO inside the flow model 5–6 min per run at four function evaluations, 20–26 min with the mean of eight; PPO in the real environment 3.7–7.7 min on pixels with the convolutional torso and 51–70 s with the MLP torso, 76–78 s on the frozen latent, 4.7–4.8 min on the decoded latent with the convolutional torso and 1.8–2.9 min with the MLP torso; the latent diagnostic: 2 min.

E Evaluation protocol

Every gate-policy endpoint uses the final checkpoint and 256 real episodes from reset, with the 128-step cap. Survival is the fraction still alive after step 128; death on that step is not survival. Walls passed counts wall exits before or on the terminal transition, including an exit on the same step as death. Discovery is final survival strictly above 50%. Real-training and imagined-training runs advance their random keys under different iteration schedules; their final evaluations are not paired episode trials. Policy seeds are separate runs conditional on a fitted model; independently fitted encoders provide a different replication level. The flow model has only one checkpoint. Reported means weight policy runs equally; standard deviations describe variation across those runs, not uncertainty across model families.

Gate specification. The gate uses the same image and object sizes and cardinal action displacement. A white vertical wall of width three pixels spawns at $x = 60$ every 22 steps and moves left three pixels per step. Its 16-pixel gap is sampled at spawn, top or bottom with equal probability, and becomes visible when the wall reaches $x \leq 40$. The gap location is hidden state before that reveal. Border or wall contact produces reward -1 and termination; the time cap is 128 steps.

Latent diagnostic. For each encoder, the diagnostic uses 64 real contexts with the wall still solid at $x \in \{42, 45\}$. Reconstructed simulator states generate recorded futures and matched wall-free futures under the recorded actions. The recorded branch survives the diagnostic horizon by construction. At each step, wall amplitude is the squared Euclidean distance between encodings of the wall-present and wall-absent frames. The teacher-forced residual uses real encoded history and is a total squared error, summed over latent coordinates. The wall coordinate of a prediction is its projection onto the wall-present minus wall-absent latent difference, with endpoints 0 and 1. Decoded wall intensity averages the minimum colour channel on shared wall pixels, excluding either possible gap band. These measurements compare features within one encoder; raw distances across models are not an information metric.

F Latent-policy torso controls

The MLP on decoded latents passes 0.92 walls per episode with no discoveries in six runs (Table 5), using encoders 0 and 1. The decoded CNN control instead uses all three encoders, with three

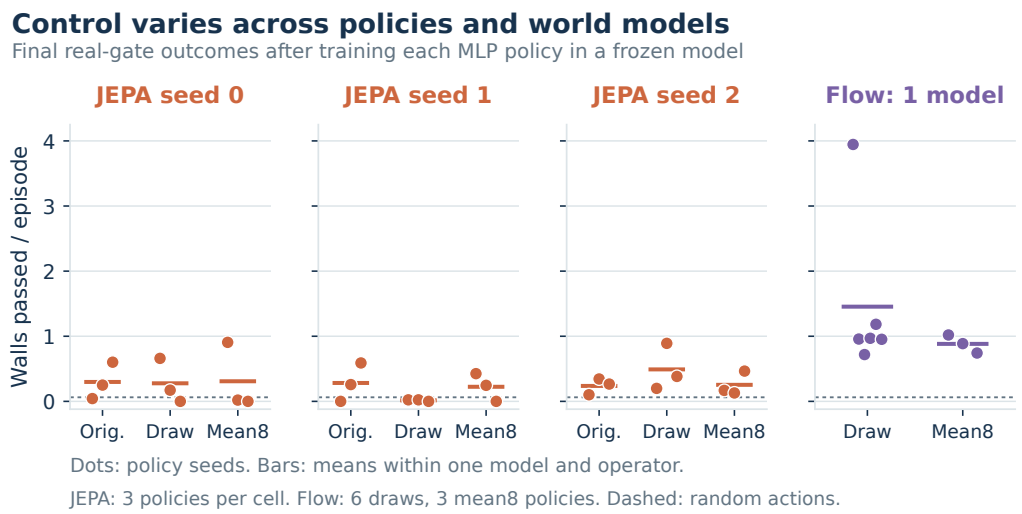


Figure 4: **Control outcomes by model fit and predictor variant.** Each point is a final policy evaluated on 256 real episodes. Original prediction, residual sampling, and residual means are separate conditions; model and policy replication are distinct. Horizontal bars show within-model means, and the dashed line marks random actions. The flow reference uses one checkpoint.

policies per encoder. Both of its discoveries use encoder 1. The decoder is frozen and reconstruction-supervised; success therefore establishes useful encoded information under this added read-out, not sufficiency of the unsupervised training objective. Failure of the true-pixel MLP recipe establishes no upper bound on MLP capacity.

Table 5: Real-gate observation and torso controls: final walls per episode and survival, mean ± standard deviation over policies; discovery means final survival above 50%. All policies use 1,499,136 real transitions.

observation of the real-gate PPO	runs	walls passed	survival (%)	discovered
pixels, convolutional torso	3	2.91±1.65	65±52	2/3
pixels, MLP torso	3	0.91±0.08	4±2	0/3
frozen JEPA latent, MLP torso	9	0.94±0.04	4±1	0/9
decoded JEPA latent, convolutional torso	9	1.57±1.26	22±37	2/9
decoded JEPA latent, MLP torso	6	0.92±0.08	3±2	0/6

G CONTROL OUTCOMES AND LATENT DIAGNOSTICS

Table 6 gives aggregate final outcomes; Table 7 lists every imagined JEPA policy. The pooled imagined JEPA mean weights all original, sampled-residual and mean-residual runs equally. The flow rows distinguish six sampled policies from three policies using a mean of eight, all sharing the same model fit. They do not estimate variation across independent flow-model fits. Tables 8 and 9 retain the encoder and geometry diagnostics supporting the gate observations.

Table 6: Final control outcomes in the real gate, mean ± standard deviation over policy runs. Imagined training uses 1,228,800 transitions per policy; real training uses 1,499,136. Model-fit counts and policy-run counts identify distinct replication levels.

Controller	Models	Policies	Walls	Survival (%)	Discovery
JEPA: original	3	9	0.27 ± 0.22	0 ± 1	0/9
JEPA: mean of 8	3	9	0.26 ± 0.30	1 ± 1	0/9
JEPA: one sample	3	9	0.26 ± 0.32	1 ± 1	0/9
Flow: mean of 8	1	3	0.88 ± 0.14	3 ± 1	0/3
Flow: one sample	1	6	1.45 ± 1.23	19 ± 39	1/6
Random	–	–	0.06	0	–
Alternating	–	–	0.02	0	–
Real pixels: CNN	–	3	2.91 ± 1.65	65 ± 52	2/3
Real pixels: MLP	–	3	0.91 ± 0.08	4 ± 2	0/3
Real JEPA: MLP	3	9	0.94 ± 0.04	4 ± 1	0/9
Real decoded: CNN	3	9	1.57 ± 1.26	22 ± 37	2/9

Table 7: Every imagined JEPA policy at its final checkpoint: survival (percent) and walls per episode by encoder seed, predictor variant and PPO seed. Each cell uses 256 real episodes.

		Survival (%)			Walls passed		
Predictor	Model	PPO 0	PPO 1	PPO 2	PPO 0	PPO 1	PPO 2
Original	0	0	0	1	0.04	0.25	0.60
	1	0	0	1	0.00	0.26	0.59
	2	0	1	0	0.10	0.34	0.27
Residual mean	0	3	0	0	0.91	0.02	0.00
	1	2	1	0	0.43	0.25	0.00
	2	0	0	1	0.17	0.13	0.46
Residual sample	0	2	1	0	0.66	0.17	0.00
	1	0	0	0	0.02	0.02	0.00
	2	0	3	1	0.20	0.89	0.38

Table 8: Grid-JEPA diagnostics by encoder seed. Wall amplitude and total teacher-forced residual are squared distances at future steps 1 and 8; decoded intensity uses steps 1 and 2. The wall coordinate has endpoints 0 (wall absent) and 1 (true encoding). Head AUC compares each death cause with alive transitions; PPO wall counts pool predictor variants.

JEPA encoder seed	0	1	2
Wall amplitude, step 1	0.035	0.028	0.088
Wall amplitude, step 8	0.124	0.082	0.193
Total residual, step 1	0.145	0.130	0.156
Total residual, step 8	0.249	0.228	0.279
Wall coordinate, step 1	0.39	0.45	0.66
Decoded wall, step 1	0.00	0.00	0.68
Decoded wall, step 2	0.00	0.00	0.02
Border-death AUC	0.96	0.96	0.88
Wall-death AUC	0.65	0.65	0.76
Imagined PPO, walls	0.29±0.34	0.17±0.22	0.33±0.24

Table 9: JEPA geometry and decoded wall intensity for model seed 0, averaged over 64 pre-reveal contexts. Distances use matched actions and the same encoder; total residuals include all latent coordinates. Wall intensity is measured on shared wall pixels.

Future step	1	3	5	8
Squared distance, summed over coordinates				
Wall-present versus wall-absent	0.035	0.041	0.071	0.124
Total teacher-forced residual	0.145	0.152	0.200	0.249
Residual along wall direction	0.015	0.021	0.048	0.104
Residual orthogonal to wall direction	0.130	0.132	0.152	0.145
Predicted wall coordinate	0.39	0.33	0.29	0.21
Decoded wall intensity				
Encoded true future	1.00	1.00	1.00	1.00
Original JEPA rollout	0.00	0.00	0.00	0.00

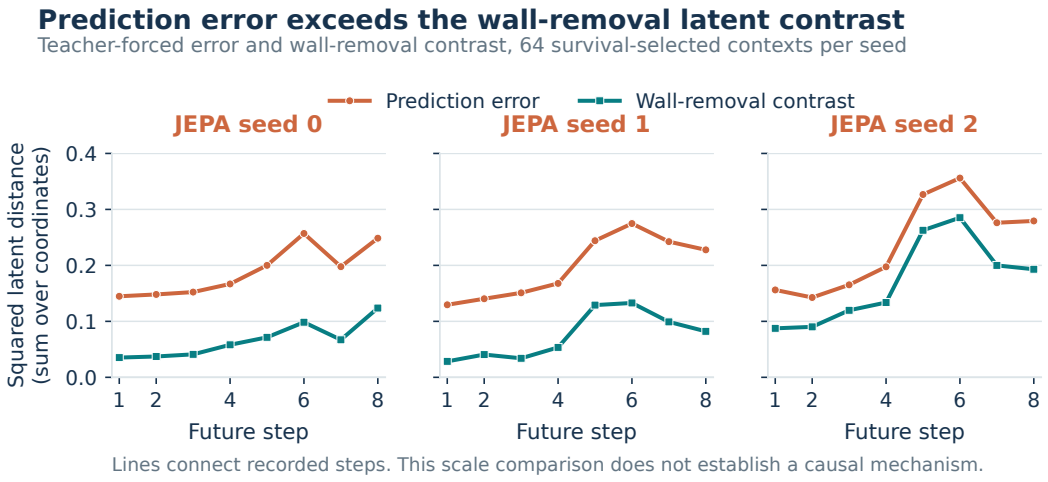


Figure 5: **Within-encoder scale diagnostics.** For each of the three gate encoders, wall-removal distance and teacher-forced total one-step residual are summed squared latent distances. Curves average 64 pre-reveal contexts whose recorded futures survive the diagnostic horizon. These are feature sensitivity and total prediction error, not commensurate measures of information content.

H TERMINAL REWARDS IN THE FLOW RECIPE

The original recipe masks transitions entering reset frames out of reward training. Here the only nonzero reward occurs on terminal transitions, so that mask removes the reward signal. The corrected head predicts the next reward from the current frame summary and action with terminal transitions retained in its loss. The reported flow-policy outcomes use this corrected convention. The single-checkpoint reference therefore denotes the complete tokenizer, dynamics, reward and termination package, rather than an isolated change in transition stochasticity.

I CRAFTAX PROTOCOLS AND SINGLE-CHECKPOINT RESULTS

I.1 FRESH LOCAL-GOAL EVALUATION

The new evaluation uses the same existing 512-dimensional checkpoint, model seed 0, with 128 starts for each evaluation seed 101, 202, and 303. Goal generation and real execution have separate pseudorandom streams; all controllers share initial states and the execution stream. The target is the endpoint after six random actions, following seven warm-up actions from reset. The objective is final player-position equality with survival throughout the six executed actions. Every evaluated trajectory survives, so the survival requirement does not alter the measured rates. The target position displacements are zero, one, two, or three grid cells in 137, 148, 84, and 15 cases, respectively.

The open-loop planner solves once at horizon six. Receding-horizon planning receives each real observation, executes the first selected action, and replans over the remaining horizons $6, 5, 4, 3, 2, 1$. Every CEM solve uses 512 candidate sequences, 32 elites, and six iterations. Thus receding planning uses six times as many candidate-sequence searches and 3.5 times as many predicted latent transitions as one-shot planning, excluding additional encoding. It is not a compute-matched feedback intervention. Both controllers use terminal latent MSE and the native 17 actions, without reward heads, decoder use, latent rescaling, or privileged coordinates in the planning cost.

Table 10: Craftax goal-success counts and Wilson 95% intervals, conditional on one fitted model. The reference replays the privileged sequence used to generate the goal under independent execution noise; it is a protocol check, not an oracle or a guaranteed upper bound.

Controller	Success / total	Rate (%)	Wilson interval (%)
No-op	$137/384$	35.68	$[31.05, 40.59]$
Random	$65/384$	16.93	$[13.51, 21.00]$
JEPA open loop	$358/384$	93.23	$[90.26, 95.34]$
JEPA receding horizon	$369/384$	96.09	$[93.66, 97.62]$
Reference-action replay	$382/384$	99.48	$[98.12, 99.86]$

Open-loop and receding-horizon success on displaced targets is $222/247$ and $232/247$; for displacements of at least two cells it is $80/99$ and $87/99$. Random actions reach $19/247$ and $1/99$ in these strata, while no-op reaches neither. Mean endpoint position L1 errors are 0.1068 open loop, 0.0781 receding horizon, 1.5599 random, and 0.9401 no-op, in grid-cell units. Receding-horizon success exceeds open loop by 2.86 percentage points; a paired episode bootstrap with 10,000 resamples gives $[0.78, 4.95]$ percentage points. All intervals quantify evaluation variation for this fixed checkpoint. The three evaluation seeds are not training replications.

Table 11: Craftax success by target displacement. The first column includes all 384 starts; the others retain 247 displaced targets and 99 targets at least two grid cells away.

Controller	All (384)	Displaced (247)	≥ 2 cells (99)
No-op	35.7%	0.0%	0.0%
Random	16.9%	7.7%	1.0%
JEPA open loop	93.2%	89.9%	80.8%
JEPA receding horizon	96.1%	93.9%	87.9%

I.2 Earlier exploratory goal and reward protocols

The earlier results below use different evaluation episodes and a shared goal-generation/execution noise protocol. They are not pooled with the fresh independent-noise evaluation above.

The grid checkpoint uses training seed 0, a tile encoder, $16 \times 32 = 512$ latent coordinates, a six-layer factored predictor of width 256, history length eight, 17 actions, batch 128, and 8,500 updates. The recorded parameter count is 11.12 million. Training takes 604.5 seconds from loading through the logged end-to-end timer (578.5 seconds in the update loop); thus it is an approximately ten-minute run, not a strict sub-600-second result. The frozen decoder and heads add 175.6 seconds. Results are from one model fit and evaluation seed 0.

Goal generation and planning. Reset states receive seven random warm-up actions. Six further random actions generate a reachable goal image. CEM optimizes a length-six action sequence in latent space, with population 512, 32 elites, and six iterations. The learned planner uses the goal image, without privileged position coordinates or the generating actions in its cost. It executes the entire selected sequence without observation feedback. Success is equality of final player positions. Goal generation, random actions, and the selected plan use the same future simulator random keys; replaying the generating actions therefore reaches all goals and serves only as a protocol check. This protocol samples nearby reachable goals, not arbitrary distant tasks.

The planned endpoint position L1 error is 0.0781, against 1.5313 for random actions. Pixel MSE on the 0–255 scale is 104.00 versus 1477.92. These quantities use the same 128 starts as the reported $121/128$ versus $24/128$ reach counts. They do not quantify variation across fitted models.

Reward control and head generalization. Reward and value heads use the first 80,000 transitions (320 time rows across 250 streams) from the world-model training archive. Held-out time rows are excluded from head fitting but not from encoder training. Value fitting uses the first 80 time rows owing to truncated-return handling. Reward and value fitting correlations are 0.922 and 0.995; held-out correlations are 0.353 and 0.266. Held-out reward RMSE is 0.222. Mean predicted held-out value is 2.356, versus mean empirical discounted return 1.348.

Reward evaluation uses 128 starts, 100 steps, no autoreset, and rewards accumulated while alive. The CEM controller executes one action and replans from each real observation. Random actions, myopic fitted-reward argmax, one predicted step plus value, and CEM obtain returns 1.4453, 0.7813, 0.9438, and 0.4734, respectively; their across-start standard deviations are 1.0897, 1.1109, 1.2014, and 1.1039. Fractions alive at the end are 0.6328, 0.7266, 0.6328, and 0.6719. Aggregate summaries do not preserve the per-start covariance needed for paired confidence intervals. The archive's mean reward multiplied by 100 is not an online behavior baseline and is excluded from the comparison.

Earlier Craftax experiments used a different 81×32 spatial latent and different horizons. They are not merged with these 512-dimensional results. There is no matched online flow-controller comparison for this checkpoint.

J Earlier unmatched gate experiments

Reference model. For the gate, a reconstruction-trained tokenizer and causal flow-matching predictor provide a feasibility reference (Lipman et al., 2023; Chen et al., 2024; Hafner et al., 2025). It uses the same latent shape, data, and nominal model-training budget, with a 32-frame context and jointly fitted reward and termination heads. Architecture, supervision, and history length differ from the JEPA. One dynamics checkpoint is evaluated, so its policy runs show variation conditional on that fit rather than variation across fitted flow models.

J.1 Useful wall information remains in the frozen representation

Decoded real encodings visibly retain the approaching wall (Figure 6). To test whether that information can support behavior, we train a convolutional policy in the real gate on the frozen decoder's output. Two of nine runs discover the survival strategy, both using encoder seed 1; the corresponding real-pixel CNN discovers it in two of three runs (Figure 7). This establishes useful information in at least one fitted encoder under the decoder-and-CNN read-out. The decoder adds reconstruction

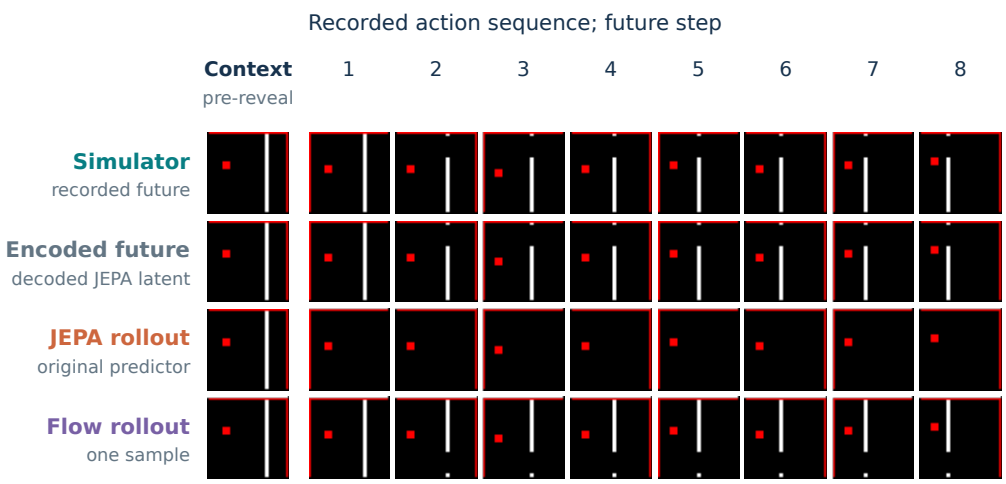


Figure 6: **A wall visible in real encodings disappears in predicted rollouts.** A pre-reveal context is followed by futures under recorded actions: the simulator, decoded real JEPA encodings, decoded original JEPA predictions, and one flow-model sample. The JEPA rows use encoder seed 0 and the same fitted decoder. The example is the first stored diagnostic context at wall position $x = 45$ among 64 contexts whose recorded futures survive the horizon. The flow sample is a single-checkpoint feasibility reference. A decoding failure on predicted latents can reflect both prediction error and off-distribution read-out; it is not alone proof of information loss.

supervision outside the world-model budget, so this is a test of recoverable information rather than unsupervised control sufficiency.

The policy interface matters. With an MLP torso, real pixels yield 0.91 walls per episode and frozen latents yield 0.94, with no discoveries in either group. An MLP on decoded latents similarly yields 0.92 walls. Thus failure of this MLP training recipe is not evidence that the encoder has lost the obstacle. These comparisons do not set a capacity limit for all MLP policies; they locate a limitation of the tested read-out and optimization procedure.

J.2 The tested imagination pipeline does not discover the strategy

In the earlier unmatched protocols, all three JEPA predictor variants produce poor downstream control (Table 12; per-seed outcomes in Appendix G). The original predictor yields 0.27 walls per episode; one residual sample and the mean of eight each yield 0.26. No policy among these 27 runs discovers the strategy; the highest final survival is 3.1%. Real-transition PPO on the same frozen encoders yields 0.94 walls. The difference measures the combined effect of learned transitions, fitted heads, and the changed PPO protocol. Even the stronger real-latent MLP result falls far short of reliable survival.

The flow reference supports feasibility at this task and nominal model budget: one of its nine policies discovers the strategy. The remaining eight pass 0.72–1.18 walls per episode. Since all nine use one dynamics fit with a different architecture and supervision, this reference does not establish a model-family ranking. Training the gate JEPA for twice and four times the steps also leaves imagined-policy survival at 0–3% in the tested extensions (encoder seed 0, policy seed 0, each predictor variant).

J.3 Prediction and fitted heads provide distinct failure signals

The same decoder that renders real encoded walls removes them from original predicted rollouts. Across all three encoder seeds, decoded shared-wall intensity is at most 0.02 by the second predicted step (Appendix G). The shared wall pixels occur in both possible gap futures, so a pixelwise mixture of those futures would retain them. This observation rules out that simple pixel-space picture; it does not determine what averaging or distribution shift does in the learned latent space.

Table 12: **Final gate outcomes, with predictor variants kept separate.** Mean ± sample standard deviation across policy runs; discovery is final survival above 50%. JEPA imagination variants and real-latent conditions each contain 3 models × 3 policies; pixel controls have three policies. Real training changes rewards, terminations, and PPO protocol alongside dynamics. Flow is a single-model reference.

Training / observation	Predictor or torso	Walls	Discovery
Imagined JEPA	Original	0.27 ± 0.22	$0/9$
	Residual sample	0.26 ± 0.32	$0/9$
	Residual mean of 8	0.26 ± 0.30	$0/9$
Real / frozen JEPA	MLP	0.94 ± 0.04	$0/9$
Real / pixels	MLP	0.91 ± 0.08	$0/3$
Real / decoded JEPA	CNN	1.57 ± 1.26	$2/9$
Real / pixels	CNN	2.91 ± 1.65	$2/3$
Imagined flow	Sample	1.45 ± 1.23	$1/6$
	Mean of 8	0.88 ± 0.14	$0/3$

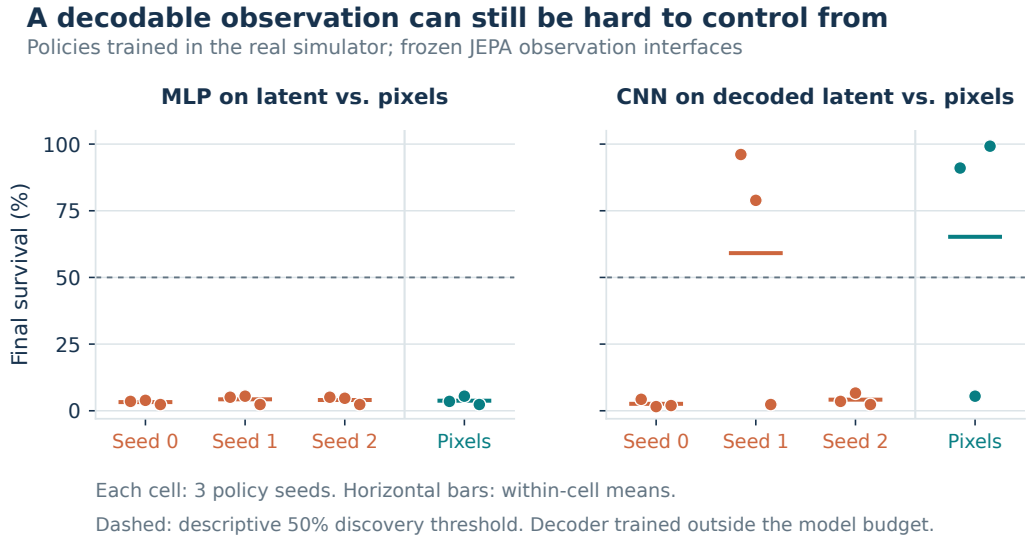


Figure 7: **Recoverable information can support successful control.** Final real-gate survival for MLP policies on frozen latents or pixels (left), and CNN policies on decoded latents or pixels (right). Points are policy runs, grouped by encoder seed; the dashed line is the 50% discovery criterion. Both decoded-latent discoveries use encoder 1. Reconstruction supervision and CNN training are additional to representation fitting.

A within-encoder counterfactual diagnostic makes the scale of the feature explicit. On 64 pre-reveal contexts for seed 0, the first-step squared distance between wall-present and wall-absent encodings is 0.035, while total teacher-forced prediction error is 0.145. The same ordering holds for all three seeds (Appendix G). This comparison concerns sensitivity to wall removal versus error in all features. It neither identifies a causal mechanism nor permits comparing information content through raw distances across independently fitted latent spaces.

The fitted heads expose another limitation. For encoder 0, termination AUC is 0.96 for border deaths but 0.65 for wall deaths. Predicted rewards for the two death causes are -0.89 and -0.05, respectively, against the true -1. An imagined policy therefore trains on a weak wall-collision signal. The residual-sampling variants retain these heads and the policy protocol. Their poor control outcomes therefore test this particular intervention, not distributional prediction in general.

K MATCHED-CONTROL PROTOCOL AND OUTCOMES

The primary matrix contains three fixed gate model checkpoints, five policy seeds (0–4), and three transition/signal conditions. Within each model/policy block, initialization, sampled starts, source identities, checkpoint hash, data hash, requested updates, policy parameter count, evaluation seed, and episode count must match. Start arrays are compared by semantic content hashes. Only complete three-condition blocks with 150 finished updates, finite required training diagnostics, and scalar outcomes reproduced from saved episode arrays enter the complete analysis. Missing, partial, or invalid runs retain their place in the planned denominator.

The primary endpoints are wall exits before or on the terminal transition and survival through the 128-step cap. A same-step wall exit counts even if that transition is terminal. A death on step 128 is counted as death, using episode return rather than duration. Survival above 50% is a secondary operational threshold, not independent evidence of a specific learned maneuver. Each saved final policy is evaluated on 256 real episodes with a common key schedule. Outcomes are paired within model and policy seed. Raw hidden state is used only to reconstruct the simulator transition source and must reproduce sampled current pixels exactly before training proceeds.

Within a fixed model, policy-seed effects are equally weighted. The main descriptive interval uses 10,000 paired-policy bootstrap resamples with analysis seed 7341, conditional on the evaluation set. A separate interval resamples evaluation slots jointly across policies while holding fitted policies fixed. It measures evaluation variation, not policy-training uncertainty. Across model fits, we show all three effects, their equal-model mean, and range; we do not treat 15 policy pairs as 15 independent models or report a model-family interval. All intervals are unadjusted and descriptive.

Exact-simulator states become absorbing after true termination. When fitted signals are used, their learned continuation can remain positive there; post-terminal alive mass and weighted reward are reported as part of the intervention's support change.

K.1 COMPLETED PRIMARY OUTCOMES

All 45 prescribed MLP runs passed full-budget, pairing, checkpoint-identity, and per-episode outcome checks. Their primary outcomes appear in Table 13. Table 14 retains every predeclared wall-count contrast with its within-model policy-seed interval. No confidence interval here estimates variation over a model family.

Table 13: **The exact training environment does not make this PPO recipe solve the gate.** Final real-environment outcomes, mean $\pm$ sample standard deviation across five policy seeds per row. Each of three fitted models receives all three matched conditions. None of the 45 policies exceeds 50% survival. "Fitted" means the unchanged learned reward and termination heads; "true" means simulator signals.

Model	Transitions / signals	Walls per episode	Survival (%)
0	Learned / fitted	0.285 ± 0.265	0.94 ± 1.28
0	Exact / fitted	0.230 ± 0.366	0.62 ± 0.98
0	Exact / true	0.452 ± 0.099	0.47 ± 0.33
1	Learned / fitted	0.217 ± 0.356	0.78 ± 1.35
1	Exact / fitted	0.068 ± 0.152	0.16 ± 0.35
1	Exact / true	0.341 ± 0.210	0.39 ± 0.39
2	Learned / fitted	0.320 ± 0.196	0.39 ± 0.28
2	Exact / fitted	0.096 ± 0.119	0.00 ± 0.00
2	Exact / true	0.543 ± 0.119	1.41 ± 0.45

L NATIVE-GRID POLICY SENSITIVITY

The sensitivity uses the same frozen models, rollout protocol, and three transition/signal conditions, with three policy seeds per model. The policy applies a CNN to the existing $4 \times 4 \times 32$ topology

Table 14: Matched primary wall-count differences. Intervals are 95% percentile bootstraps over five paired policy seeds within one fixed model, conditional on its evaluation set. They are unadjusted descriptive intervals; all three prespecified contrasts are retained.

Model	Contrast	Δ walls	95% interval
0	Exact/fitted − learned/fitted	-0.055	$[-0.368, +0.259]$
0	Exact/true − exact/fitted	$+0.222$	$[-0.109, +0.452]$
0	Exact/true − learned/fitted	$+0.166$	$[+0.015, +0.348]$
1	Exact/fitted − learned/fitted	-0.149	$[-0.492, +0.045]$
1	Exact/true − exact/fitted	$+0.273$	$[-0.027, +0.498]$
1	Exact/true − learned/fitted	$+0.123$	$[-0.195, +0.437]$
2	Exact/fitted − learned/fitted	-0.223	$[-0.429, +0.046]$
2	Exact/true − exact/fitted	$+0.447$	$[+0.329, +0.598]$
2	Exact/true − learned/fitted	$+0.223$	$[+0.034, +0.420]$

and has 583,236 parameters. All 27 runs complete 150 updates and 1,228,800 requested transitions. Nine three-arm start-array pairs, model identities, frozen source and policy hashes, and every scalar endpoint are verified from saved artifacts.

Table 15: Native-grid CNN sensitivity, kept separate from the primary MLP campaign. Each cell is mean $\pm$ sample standard deviation across three policies from one fixed model. None of the 27 policies exceeds 50% survival.

Model	Transitions / signals	Walls per episode	Survival (%)
0	Learned / fitted	0.000 ± 0.000	0.00 ± 0.00
0	Exact / fitted	0.431 ± 0.436	2.08 ± 1.85
0	Exact / true	0.616 ± 0.090	1.17 ± 0.39
1	Learned / fitted	0.284 ± 0.246	0.65 ± 0.60
1	Exact / fitted	0.738 ± 0.067	2.34 ± 0.68
1	Exact / true	0.367 ± 0.238	0.52 ± 0.45
2	Learned / fitted	0.202 ± 0.158	0.00 ± 0.00
2	Exact / fitted	0.229 ± 0.370	0.39 ± 0.68
2	Exact / true	0.452 ± 0.256	0.65 ± 0.81

Within-model exact/true minus learned/fitted wall-count effects are 0.616, 0.083, and 0.250; the corresponding paired-policy bootstrap intervals are $[0.516, 0.691]$, $[-0.312, 0.395]$, and $[-0.148, 0.477]$. These coarse intervals use only three paired policy seeds and condition on each checkpoint and its fixed evaluation set. They are not pooled with primary MLP intervals.

M FORECAST-READOUT PROTOCOL AND OUTCOMES

The three gate models stay frozen. Up to 16,384 training contexts and 8,192 evaluation contexts are sampled without replacement with seeds 20260919 and 20260920, using matching context IDs across models. Past context may not cross a reset. Forecasts use the original recorded outgoing actions, with horizons 1, 2, 4, and 8. Parameters and BatchNorm statistics must remain unchanged during extraction, and cached encodings are checked against the public rollout implementation.

The three feature domains are real future encoding, autoregressive forecast, and copied current encoding. Each domain has 512 coordinates. An affine ridge read-out uses mean squared error plus $10^{-3}\|W\|^2$, with unpenalized intercept and no hyperparameter search. Both declared scaler analyses are retained: current-training-latent mean/standard deviation shared across domains; and source-domain training mean/standard deviation for each horizon. Both use a fixed 10^{-6} scale floor, and every transfer applies its source-fitted scaler. No evaluation data determine a scaler, regularizer, checkpoint, or reporting condition.

Cube coordinates are divided by 63 for fitting and reported in pixels. Wall position is fitted and evaluated only on truth-present rows; the absent-wall sentinel is never regressed as a coordinate. Presence reports class prevalence, AUROC, average precision, and threshold accuracy/balanced accuracy, with undefined one-class metrics omitted. Every task includes a constant training-label-mean baseline and sample counts. A task with fewer than 64 fitting rows is marked insufficient.

The archive stores reset frames after terminal transitions, so physical-state read-outs use a horizon-specific continuing-episode mask. In particular, one-step evaluation does not select for eight-step survival. The hidden-gap negative control instead retains every pre-reveal context, including later resets, because future-survival selection could reveal the gap through the recorded actions. The gap label refers to the original current wall. Chance forecasting of this unavailable bit is compatible with correct treatment of uncertainty.

All fitted weights, scalers, selected IDs, labels, masks, per-example predictions, and provenance hashes are saved. Overlapping windows are not independent samples; any uncertainty uses evaluation stream or episode clusters. A weak read-out remains consistent with information loss, poor conditioning, limited support, or insufficient probe capacity.

Execution and uncertainty. All three models use the same canonical padded frame batch and predictor batch. A strict parity check against the public rollout function is bit-exact at the production shape, and model parameters and BatchNorm statistics remain unchanged. This canonicalization was fixed after an initial smoke parity failure and before any successful read-out fit; no tolerance, model weight, or metric was changed. The analysis recomputes 1,080 metric rows and 720 paired contrasts from saved float32 predictions without refitting. Intervals resample the 100 evaluation environment streams jointly across compared read-outs, with 2,000 replicates and seed 20260921. They condition on the fitted model and probe. AUROC and average precision are reported as point estimates.

For the model-2, horizon-eight pre-reveal cohort, the forecast-versus-copy MAE difference is -0.63 pixels with interval $[-1.41, 0.13]$ under source-specific scaling. The transfer-versus-adapted difference is 3.85 pixels with interval $[3.48, 4.22]$. Common scaling gives -0.64 $[-1.42, 0.12]$ and 3.89 $[3.52, 4.25]$, respectively. These are descriptive conditional comparisons; the strong adaptation result occurs in one of the three fitted models.

M.1 Read-out errors and a post-hoc tracking audit

The predeclared absolute-error table is retained below for every fitted model and both scaling analyses. Its global training-mean baseline is evaluated on the selected pre-reveal slice; it is not a slice-specific optimal constant. A separate audit of saved predictions examined correlations and RMSE without refitting or modifying any prediction. The model-2 forecast-fitted probe has Pearson correlation 0.246 on the full wall-present cohort and -0.175 within the pre-reveal slice. Its full-cohort RMSE is 18.65 pixels versus 13.49 for the global training-mean constant. Thus the MAE improvement should not be described as reliable tracking or successful recovery of wall position. A subsequent train-only constant-baseline audit uses the same horizon and cohort masks, with 1,509 training and 716 evaluation rows in the pre-reveal slice. The training-cohort mean position is 29.433 pixels and the median is 30. Their evaluation MAEs are 4.769 and 4.642 pixels, respectively, improving on every highlighted model-2 feature read-out in that slice. The full-cohort training median has MAE 10.560, compared with 9.112 for the forecast-fitted probe, while the latter's RMSE remains worse than the mean constant. This is an explicitly post-hoc interpretation check, with no feature-probe refitting, clipping, latent rescaling, or alteration of the declared metric outputs.

N Triggered termination sensitivity

This follow-up was triggered by the primary experiment's monitored post-terminal weights. It adds one condition: exact transitions, true termination, and the original learned reward evaluated on the current latent and outgoing action. It reuses every corresponding reference arm's saved start indices, checkpoint and data identity, policy initialization, update budget, and evaluation keys. The MLP stratum adds five policy seeds for each of three models; the native-grid stratum adds three, for 24 additional runs. The original 72 policies are neither retrained nor relabeled.

Table 16: **Read-out fitting changes measured forecast error.** Eight-step wall-position MAE in pixels on 716 pre-reveal continuing contexts. R, F, and C denote real future, forecast, and copied-current latents; arrows indicate fitting and evaluation domains. Both train-only scaling analyses are shown. The global training-mean baseline has MAE 14.24; the post-hoc training-cohort mean and median have MAE 4.77 and 4.64. The larger refitting gain occurs in model 2; all three models are retained.

Model	Scaling	R→R	R→F	F→F	C→C
0	Common	11.66	13.79	13.99	10.77
0	Source	11.37	13.77	13.99	10.77
1	Common	13.00	14.21	13.93	11.28
1	Source	12.82	14.44	13.93	11.28
2	Common	6.62	10.05	6.16	6.79
2	Source	6.53	10.02	6.16	6.79

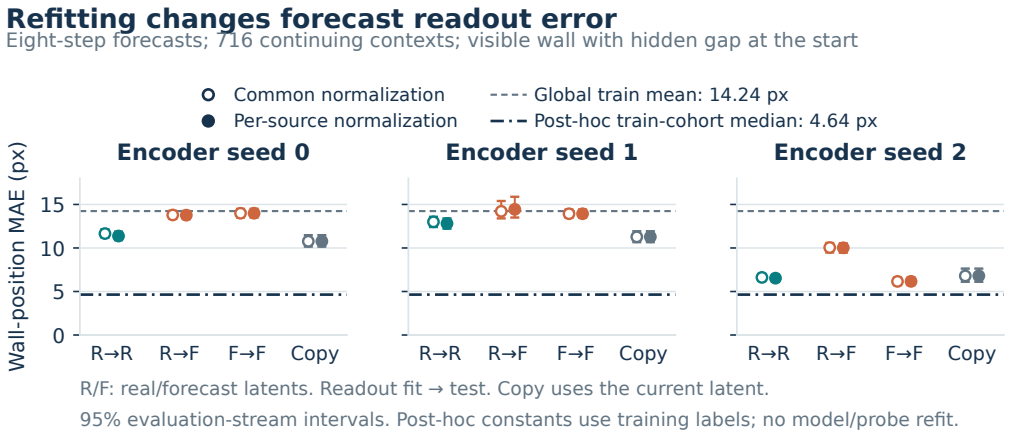


Figure 8: **Refitting changes forecast read-out error.** All three model seeds and both predeclared normalization schemes are retained. The original baseline is the global training-label mean evaluated on the selected slice; the added cohort-median constant is a post-hoc train-only check. Lower slice MAE can reflect calibration as well as tracking; the model-2 correlation audit and full-cohort RMSE prevent a reliable-position-recovery claim.

All added runs complete 150 updates. All 3,600 recorded training updates have finite required diagnostics and exactly zero continuation-weighted post-terminal mass and reward. Every final policy binary hash and per-episode endpoint is checked. Table 17 gives the new condition's outcomes. None of the added policies exceeds 50% survival.

Table 17: Exact transitions, true termination, and fitted reward: the added termination-sensitivity condition. Mean ± sample standard deviation over paired policy seeds within one fixed model and torso. The two torso strata remain separate.

Torso	Model	Policies	Walls per episode	Survival (%)
MLP	0	5	0.164 ± 0.239	0.62 ± 0.90
MLP	1	5	0.216 ± 0.244	0.86 ± 1.05
MLP	2	5	0.078 ± 0.107	0.16 ± 0.21
Grid CNN	0	3	0.102 ± 0.110	0.65 ± 1.13
Grid CNN	1	3	0.000 ± 0.000	0.00 ± 0.00
Grid CNN	2	3	0.181 ± 0.313	0.65 ± 1.13

Table 18: Wall-count changes in the triggered sensitivity. "True termination" compares the new condition with exact transitions and fitted reward/termination. "True reward" compares exact transitions and true reward/termination with the new condition. Each entry gives the paired mean and its conditional 95% policy-seed bootstrap interval.

Torso	Model	True termination	True reward
MLP	0	-0.066 $[-0.486, +0.261]$	$+0.287$ $[+0.003, +0.512]$
MLP	1	$+0.148$ $[-0.083, +0.380]$	$+0.124$ $[-0.075, +0.391]$
MLP	2	-0.018 $[-0.162, +0.094]$	$+0.465$ $[+0.312, +0.619]$
Grid CNN	0	-0.329 $[-0.785, +0.000]$	$+0.514$ $[+0.422, +0.605]$
Grid CNN	1	-0.738 $[-0.793, -0.664]$	$+0.367$ $[+0.117, +0.590]$
Grid CNN	2	-0.048 $[-0.656, +0.543]$	$+0.271$ $[+0.164, +0.453]$

The termination contrast changes continuation weighting and removes the learned after-death extension. The reward contrast holds exact transitions and true termination fixed; policies trained under different rewards can then visit different states. The latter increases the mean wall count in all six cells, while survival changes have mixed signs. Both comparisons condition on the saved encoder/head checkpoint and the given policy/training procedure. The 10,000-resample intervals are unadjusted and descriptive; three model fits do not establish a model-family effect.